rockmoc

Method-of-Characteristics Rocket Nozzle Design in Python
Technical report and complete reference

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Abstract

rockmoc designs supersonic nozzle contours (minimum-length, ideal, truncated, Rao thrust-optimised, conical, planar NASA, annular aerospike) by the method of characteristics (MOC), and estimates their performance with an exit-plane field integral, a momentum-integral boundary layer and a JANNAF-style loss stack. Thermochemistry ranges from perfect gas through thermally perfect multi-γ isentropes to NASA CEA and Cantera. This document is the single reference for the project: theory, every public function, worked examples, all figures, validation and limits.

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1 Overview

Purpose. Education and preliminary design of rocket and test nozzles; generate CAD/CFD-ready walls (CSV, JSON, DXF, NumPy, OpenFOAM `blockMeshDict`). **Not** a CFD solver, flight-qualification tool or combustion model.

Install.

```
pip install rockmoc # core: numpy, scipy
pip install "rockmoc[plot,fast,cea]" # matplotlib, numba kernels, RocketCEA
pip install "rockmoc[all]" # + plotly, cantera, coolprop, dev tools
```

Extras: `plot`, `fast` (Numba), `viz` (Plotly), `cea`, `cantera`, `coolprop`, `dev`, `all`. Python $\geq$ 3.10, MIT licence.

Quick start.

```
from rockmoc import DesignSpec, PerfectGas, design_rao
spec = DesignSpec(gas=PerfectGas(1.2, 350.0), throat_radius=0.05,
                  exit_mach=3.0, n_characteristics=60)
r = design_rao(spec, length_fraction=0.8)
print(r.summary()["area_ratio"], r.metrics.length)
r.export_csv("rao.csv"); r.plot(save_dir="figs")
bl = r.with_boundary_layer(pc=7e6, Tc=3400, wall_temperature=800)
print(bl.meta["boundary_layer"]["delta_cf_friction"])
```

CLI.

```
rockmoc design -t ideal -g 1.2 --R 350 --Rt 0.05 --Me 3 -n 60 --rc 1.5 \
              --export csv,dxf,openfoam --units mm -o out --name bell
rockmoc run nozzle.toml # reproducible design from a spec file
```

Options: `-t mln|ideal|tic|rao|conical`, `-Me` or `-e/-area-ratio`, `-truncate` (TIC), `-length` (Rao, fraction of 15° cone), `-units m|mm|in`.

TOML spec (`spec_io.load_spec/build_design/export_design`):

```
[gas] gamma = 1.2 R = 350.0 # or molecular_weight = 23.7
[design] type = "ideal" throat_radius = 0.05 exit_mach = 3.0
               n_characteristics = 60 throat_rc_over_rt = 1.5
[boundary_layer] pc = 5.0e6 Tc = 3300.0 method = "integral" wall_temperature = 800
[export] formats = ["csv","dxf","openfoam"] units = "mm" out_dir = "out"
```

(One key per line in a real file; any `correct_wall_bl` keyword is accepted under `[boundary_layer]`.)

2 Theory and numerics

2.1 Assumptions

Steady, inviscid, irrotational, isentropic core; planar ($\delta = 0$) or axisymmetric ($\delta = 1$); shock-free design region; frozen or prescribed-γ composition; viscous effects added afterwards (BL, §2.6).

2.2 Thermodynamics

Perfect gas (`PerfectGas(gamma, R)`):

$$\frac{T_0}{T} = 1 + \tfrac{\gamma-1}{2} M^2, \quad \frac{A}{A^*} = \frac{1}{M}\left[\tfrac{2}{\gamma+1}\left(1 + \tfrac{\gamma-1}{2} M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}, \quad \nu = \sqrt{\tfrac{\gamma+1}{\gamma-1}}\tan^{-1}\sqrt{\tfrac{\gamma-1}{\gamma+1}(M^2-1)} - \tan^{-1}\sqrt{M^2-1}.$$

$c^* = \sqrt{RT_0/\gamma}\left(\tfrac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}}$, $C_F = \sqrt{\tfrac{2\gamma^2}{\gamma-1}\left(\tfrac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}\left(1 - (p_e/p_c)^{\frac{\gamma-1}{\gamma}}\right)} + \varepsilon\tfrac{p_e - p_a}{p_c}$. Inverses $M(A/A^*)$, $M(\nu)$ use safeguarded Newton with analytic derivatives (Numba kernels when installed).

Multi-γ (thermally perfect) isentrope (`build_multi_gamma_table`): $\gamma(M)$ linear $\gamma_c \to \gamma_t$ on $M \in [0,1]$ and $\gamma_t \to \gamma_e$ on $[1, M_{max}]$; integrated from stagnation with $c_p = \gamma R/(\gamma-1)$:

$$dh = c_p\,dT, \quad d\ln p = \frac{c_p\,dT}{RT}, \quad V^2 = 2(h_c - h), \quad \frac{A}{A^*} = \frac{(\rho V)^*}{\rho V}, \quad d\nu = \sqrt{M^2-1}\,\frac{dV}{V}.$$

ρV peaks at $M = 1$ for any such gas. Constant γ reproduces perfect gas to 10^{-7}. `VariableGammaGas` uses the table's own $c^* = p_c/(\rho V)^*$ and the exact $C_F = \frac{A_e}{A^*}\frac{p_e}{p_c}\gamma_e M_e^2 + \varepsilon\frac{p_e - p_a}{p_c}$ (from $\rho V^2 = \gamma p M^2$).

Chemistry backends: `RocketCEAGas` (CEA, freeze at throat/chamber), `CanteraGas` (HP equilibrium, frozen γ), `CoolPropGas`, `build_cea_frozen_table`. All satisfy the `MOCGas` protocol.

2.3 Method of characteristics

Characteristics $dr/dx = \tan(\theta \pm \mu)$, $\mu = \sin^{-1}(1/M)$; compatibility

$$d(\theta \mp \nu) = \pm\,\delta\,\frac{\sin\theta\sin\mu}{r\cos(\theta \pm \mu)}\,dx \quad \text{along } C^{\pm}.$$

Unit processes (`moc.mesh`): interior, axis ($\theta = 0$, reached along C^-), wall (prescribed angle or streamline). Second-order predictor–corrector on averaged θ, μ, r. Walls are streamlines carrying the throat mass flow; transition regions are Goursat problems to the uniform exit C^+.

2.4 Transonic initial line

Hall (1962), 2nd order — default for axisymmetric throats. With $R = R_c/R_t$, $\varepsilon = 1/R$, lengths/R_t, $z = x\sqrt{2R/(\gamma+1)}$:

$$u = 1 + \varepsilon u_1 + \varepsilon^2 u_2, \qquad v = \sqrt{\tfrac{(\gamma+1)\varepsilon}{2}}\,(\varepsilon v_1 + \varepsilon^2 v_2),$$

$$u_1 = \tfrac{r^2}{2} - \tfrac{1}{4} + z, \quad v_1 = \tfrac{r^3}{4} - \tfrac{r}{4} + rz, \quad u_2 = \tfrac{2\gamma+9}{24}r^4 - \tfrac{4\gamma+15}{24}r^2 + \tfrac{10\gamma+57}{288} + z(r^2 - \tfrac{5}{8}) - \tfrac{2\gamma-3}{6}z^2,$$

$$v_2 = \tfrac{\gamma+3}{9}r^5 - \tfrac{20\gamma+63}{96}r^3 + \tfrac{28\gamma+93}{288}r + z\big(\tfrac{2\gamma+9}{6}r^3 - \tfrac{4\gamma+15}{12}r\big) + rz^2.$$

Residuals of the exact potential equation fall as $\varepsilon^{3.4}$ (Sauer: $\varepsilon^{2.5}$), but the series diverges for tight throats (throat-plane axis M 1.30 at $R_c/R_t = 0.625$ vs 0.8 measured [18]). Hall is used for $R_c/R_t \geq 1.5$ (wall p/p_0 at $R_c/R_t = 2$ within 0.3% of NASA TN D-5377 vs 5% for Sauer), Sauer below. **Sauer** (first order, planar default): $\alpha = \sqrt{(1+\delta)/((\gamma+1)R_cR_t)}$, $u = \alpha x + \frac{(\gamma+1)\alpha^2 r^2}{2(1+\delta)}$. Fallback chain: Hall → Sauer → uniform near-sonic line (warned).

2.5 Contour families

MLN sharp-corner minimum-length nozzle; θ_{max} solved so the last corner C^- hits the axis at M_e (axisymmetric $\theta_{max} < \nu_e/2$; planar $= \nu_e/2$).
Ideal circular-arc throat + characteristic lattice to uniform, parallel exit.
TIC ideal contour truncated to a length fraction.
Rao TOC (`design_rao_control`, core `moc.rao.design_rao_moc`) control surface DE with $V\cos(\theta-\mu)/\cos\mu = \lambda$, $r^\delta \rho V^2 \sin^2\theta\tan\mu = K$, mass balance $\dot m_{DE} = \dot m_{DN}$; D chosen for $r_E = r_e$, θ_N for $L = fL_{15}$. *v0.8 fix:* (θ, M) on DE solved by a 2×2 Newton; the old $\theta = \mu \pm \cos^{-1}(\cdot)$ elimination was singular at $\theta = \mu$, which the optimum crosses near $f \approx 0.8$. θ_N now converges monotonically (27.47→26.95°, $n = 20 \to 120$, $\gamma = 1.2$, $\varepsilon = 13.8$).
Rao TOP `design_rao` (parabola with chart θ_N, θ_e) and `design_rao_toc` (same parabola, length scaled to the MOC ideal length; *not* a control-surface solution).
Conical 15° (or user) cone with throat arc; $\eta_{div} = (1+\cos\alpha)/2$.
Planar NASA `design_planar_mln_nasa`: TM-X-1502 sharp-throat 2-D MLN.
Aerospike Lee / Vernacchia spike (`design_spike`, truncated, base pressure, cowl).

Length ordering for equal M_e: Rao < TIC < MLN < ideal (§5, family overlay).

2.6 Boundary layer (v0.8)

`correct_wall_bl(method="integral")` solves the compressible von Kármán equation along wall arc length s with an isentropic edge state:

$$\frac{d\theta}{ds} = \frac{C_f}{2} - \theta\Big[\frac{2 + H - M_e^2}{1 + \frac{\gamma-1}{2}M_e^2}\,\frac{1}{M_e}\frac{dM_e}{ds} + \frac{\delta}{r}\frac{dr}{ds}\Big].$$

Closures: reference temperature $T^*/T_e = 1 + 0.032M^2 + 0.58(T_w/T_e - 1)$; $C_f = (\rho^*/\rho_e)\,C_{f,inc}(Re_\theta^*)$ with $C_{f,inc} = 0.0256Re_\theta^{-1/4}$ (turbulent) or $0.441/Re_\theta$ (laminar); Crocco–Busemann shape factor $H = H_iT_w/T_e + r\frac{\gamma-1}{2}M^2$ ($H_i = 9/7$, 2.59); recovery $r = Pr^{1/3}$; Bartz transport $\mu = 1.184\times10^{-7}\mathcal{M}^{0.5}T^{0.6}$, $Pr = 4\gamma/(9\gamma - 5)$. Adiabatic

wall unless `wall_temperature` given. Integrated by LSODA; on failure it falls back (with `MOCFallbackWarning`) to the zero-pressure-gradient closed form of the same closure, $\theta = [(n+1)\frac{a}{2}\frac{\rho^*}{\rho_e}(\frac{\mu^*}{\rho^* U})^n s]^{1/(n+1)}$, which equals the ODE result at $dM/ds = dr/ds = 0$. Wall offset $\delta^* = H\theta$ along the normal (TM-X-2343 rule). Friction loss $\Delta C_F = \int \tau_w \cos\alpha \, 2\pi r \, ds/(p_c A_t)$. C_f agrees with Van Driest II within 3% for $M \leq 5$, adiabatic and $T_w/T_{aw} = 0.5$.

2.7 Performance and loss stack

1-D metrics (`compute_metrics`); exit-plane field integral (`field_integrated_cf`): $F = \int(\rho u^2 + p)2\pi r \, dr - p_a A_e$, $C_F = F/(\dot{m}c^*)$ on the MOC exit ray. Integrated estimate:

$$C_F = C_{F,1D}\,\eta_{div}\,\eta_{ke}\,\eta_{chem}\,\eta_{2\phi} - \Delta C_{F,friction}.$$

η_{chem} bounds from `kinetic_efficiency_bounds`: frozen-at-throat/shifting Isp (LOX/RP-1, $O/F = 2.6$, 1000 psia, $\varepsilon = 40$: 0.945–1.0). Altitude via `pressure_at_altitude` (exponential/US-76-style to 80 km).

3 API reference

Top-level names import lazily from `rockmoc`. ⋆ = new or changed in v0.8.

Module	Function / class	Purpose
Design		
designers.base	`DesignSpec`, `.student()`, `.validate()`	inputs: gas, R_t, $M_e/\varepsilon/p_e/p_c$, n, R_c/R_t, δ
	`DesignResult`	contour, metrics, net, meta; `summary, report, lab_report, export_csv/json, with_boundary_layer⋆, explain, plot`
designers	`design_mln, design_ideal, design_tic`	MOC families
	`design_rao, design_rao_top`	Rao parabolic (TOP)
	`design_rao_toc`	TOP parabola scaled to MOC ideal length
	`design_rao_control⋆, design_rao_control_surface, rao_exit_angle`	Rao TOC by MOC control surface
	`design_conical, conical_length`	cone baseline
	`design_mln_hf, design_ideal_hf, design_tic_hf`	high-fidelity axisymmetric variants
	`design_production, ProductionResult`	fine MOC + BL⋆ + field C_F + loss stack⋆ + certificate + altitude
moc.planar_nasa	`design_planar_mln_nasa, mesh_convergence_planar`	TM-X-1502 planar MLN
moc.axisym_hf	`design_axisym_*_hf, mesh_convergence_axisym`	axisymmetric nets, convergence
moc.rao	`design_rao_moc⋆, rao_length_reference`	Rao TOC core, L_{15}
moc.ideal / kernel_mln	`build_ideal_net(s)⋆, build_mln_full_net, build_axisym_kernel`	lattices (Hall default)
moc.mesh / equations	`interior_point, axis_point, wall_point`; slopes, sources, Riemann invariants	unit processes
moc.transonic	`hall_uv, hall_line, kl_uv, kl_line, discharge_coefficient⋆, sauer_line, sauer_uv, mach_from_mstar`	throat IVL, C_d
moc.transonic_euler	`solve_throat, solve_throat_extrapolated, cuffel_wall⋆`	2-D Euler throat
moc.analysis	`analyze_wall, WallAnalysis⋆`	given-wall MOC, internal shocks

Module	Function / class	Purpose
analysis.separation	`separated_cf, psep_over_pa, SeparationResult⋆`	over-expanded separation
thermo.kinetics	`kinetic_nozzle, KineticResult⋆`	finite-rate (Cantera)
performance.two_phase	`two_phase_nozzle, TwoPhaseResult⋆`	particle lag loss
moc.types	`FlowPoint, CharNet, PointKind`	data structures
aerospike	`design_spike, design_spike_truncated, truncated_spike_with_base, estimate_base_pressure, design_primary_cowl, design_spike_with_cowl, SpikeResult`	annular aerospike
Thermo		
thermo.perfect_gas	`PerfectGas` + `area_ratio, mach_from_area_ratio, prandtl_meyer, mach_from_prandtl_meyer, mach_angle`	isentropic, PM, c^*, C_F
thermo.expansion_table	`build_perfect_gas_table, build_multi_gamma_table⋆, build_cea_frozen_table, ExpansionTable`	tabulated isentropes
thermo.variable_gamma	`VariableGammaGas⋆`	table-backed `MOCGas`
thermo backends	`RocketCEAGas, kinetic_efficiency_bounds⋆, CanteraGas, CoolPropGas`	real propellants
Analysis and performance		
analysis.boundary_layer	`correct_wall_bl⋆, two_layer_thicknesses⋆, BLCorrectionResult, bartz_viscosity, bartz_prandtl`	momentum+energy integral BL, δ^*, τ_w, ΔC_F
analysis.throat	`sauer_throat_ivl, circular_arc_throat_wall, ThroatIVL`	IVL metadata
analysis.uq	`monte_carlo_mln, sensitivity_one_at_a_time, mesh_certificate`	uncertainty, mesh certificate
analysis.convergence	`grid_convergence, richardson`	observed order, extrapolation
analysis.compare	`compare_nozzle_designs, DesignComparison`	family table
analysis.atmosphere	`pressure_at_altitude, altitude_from_pressure`	ambient
performance	`compute_metrics, integrated_performance⋆, field_integrated_cf⋆, off_design_altitude_sweep⋆, divergence_efficiency, exit_kinetic_efficiency`	C_F, Isp, loss stack
I/O, plotting, teaching, validation		
export / geometry	`export_csv, export_cad_points, export_dxf, export_blockmesh, export_numpy, Contour`	CAD/CFD output
spec_io, cli	`load_spec, build_design, export_design`; `rockmoc` CLI	reproducible runs

Module	Function / class	Purpose
plot	`plot_moc_field`, `plot_moc_field_interactive`, `plot_contour`, `plot_nozzle_contour`, `plot_wall_profiles`, `plot_wall_properties`, `plot_mach_pressure_along_wall`, `plot_contour_overlay`, `plot_design_comparison`, `plot_performance_vs_altitude`, `plot_gamma_sweep`, `plot_area_ratio_sweep`, `plot_of_ratio_sweep`, `plot_bl_displacement`, `plot_aerospike`, `plot_grid_convergence`, `display_design_result`, `display_production_result`	dark-theme figures (§5)
plot.sweeps	`altitude_sweep_rows`, `gamma_sweep_rows`, `of_ratio_sweep_rows`, `area_ratio_sweep_rows`	sweep data
student	`example_gas`, `list/describe_example_gases`, `explainer`, `format_design_report`, `generate_lab_report`, `comparison_to_markdown/latex/csv`, `KNOWN_LIMITS`	teaching aids
validate	`run_all_validation`, `run_deep_validation`, `validation_report_markdown`, `write_validation_outputs`	50 checks, 43 cited cases
exceptions	`RockmocError`, `DesignError`, `ThermoError`, `MeshError`, `ConvergenceError`, `MOCFallbackWarning`	errors carry Fix/Example tips

4 Examples

Family comparison.

```
from rockmoc import DesignSpec, PerfectGas, compare_nozzle_designs
cmp = compare_nozzle_designs(DesignSpec(gas=PerfectGas(1.2, 350), exit_mach=3, n_characteristics=40))
print(cmp.summary_table()) # L/Rt, eps, Cf_vac, eta_n per family
```

Integral BL with fallback, cooled wall, loss stack.

```
from rockmoc import correct_wall_bl, design_mln, integrated_performance
r = design_mln(spec)
bl = correct_wall_bl(r.contour, spec.gas, throat_radius=0.05, pc=7e6, Tc=3400,
                     method="integral", wall_temperature=800.0)
print(bl.meta["method"], bl.delta_star[-1], bl.delta_cf_friction)
perf = integrated_performance(bl.corrected, spec.gas, throat_radius=0.05, exit_mach=3,
                              pc=7e6, Tc=3400, delta_cf_friction=bl.delta_cf_friction,
                              eta_chemical=0.97)
print(perf.meta["loss_stack"])
```

Real propellant, kinetic bounds, variable γ.

```
from rockmoc.thermo.rocketcea_backend import RocketCEAGas
cea = RocketCEAGas("LOX", "RP1", of_ratio=2.6, Pc_psia=1000, eps=40)
eta = cea.kinetic_efficiency_bounds()["eta_estimate"]
from rockmoc import build_multi_gamma_table, VariableGammaGas
vg = VariableGammaGas(build_multi_gamma_table(gamma_c=1.25, gamma_t=1.22, gamma_e=1.18, R=350))
```

Production pipeline.

```
from rockmoc import design_production
p = design_production(spec, kind="rao", pc=7e6, Tc=3400, wall_temperature=800, eta_chemical=eta)
print(p.field.Cf_vac, p.field.meta["Cf_vac_inviscid"]); p.plot(save_dir="figs")
```

Export, convergence, UQ.

```
from rockmoc.export import export_dxf, export_blockmesh
export_dxf(r.contour, "wall.dxf", units="mm"); export_blockmesh(r.contour, "system/blockMeshDict")
from rockmoc import mesh_convergence_axisym, monte_carlo_mln
study = mesh_convergence_axisym(spec.gas, throat_radius=0.05, exit_mach=3.0, n_list=[20, 40, 80])
uq = monte_carlo_mln(throat_radius=0.05, exit_mach=3.0, gamma_mean=1.22, gamma_std=0.02, n_samples=24)
```

Teaching. `DesignSpec.student(Me=3, gas_name="lox_rp1")`, `r.report()`, `r.lab_report("lab.md")`, `r.explain()`.

5 Figures

All figures are produced by the library's plot functions. Scripts: `website/make_images.py` (site figures, `website/img`), `website/make_images_extra.py` (analysis figures), `website/make_assets.py` (dashboards and sweeps, `website/assets`). Regenerated with the v0.8 physics.

5.1 Flow fields and contours

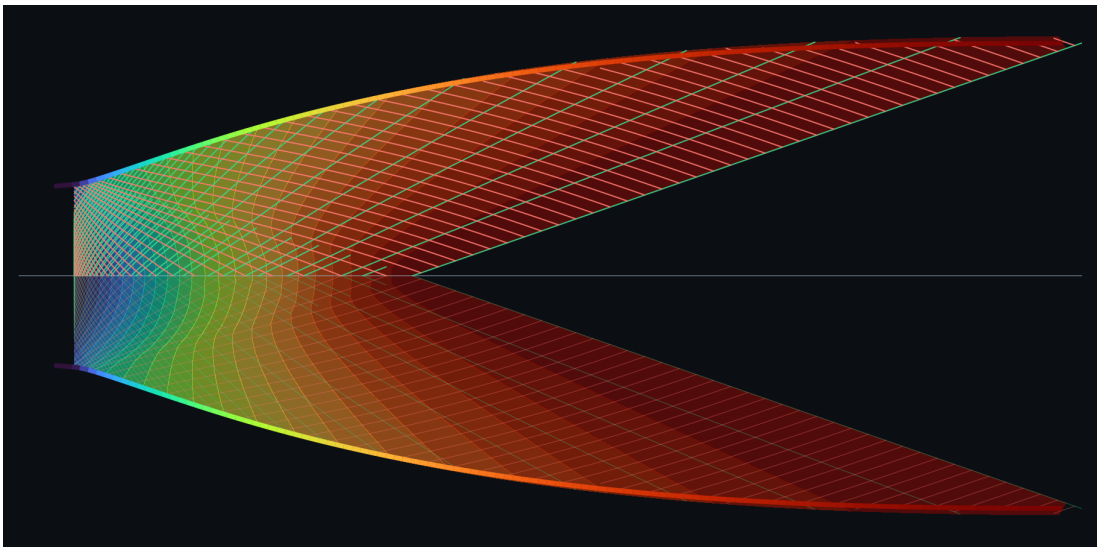
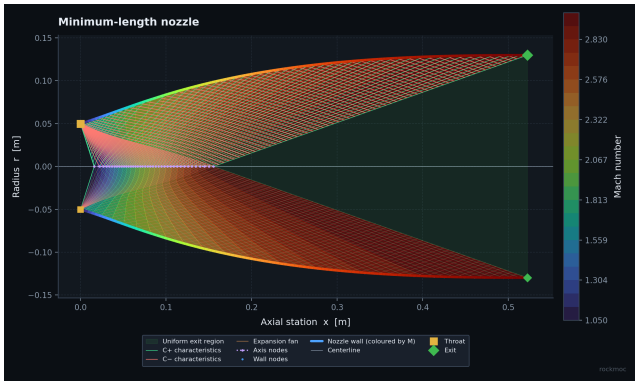
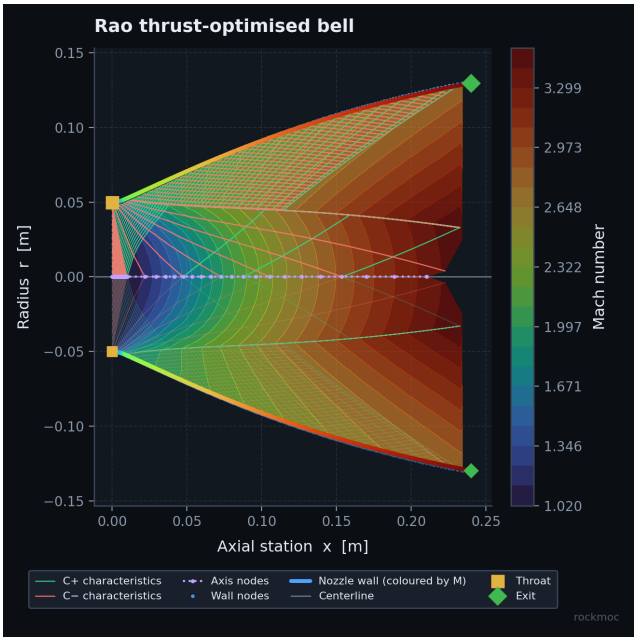


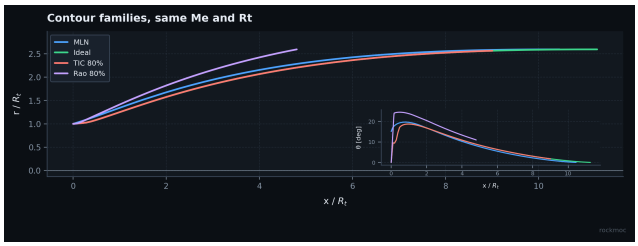
Figure 1: Hero: MOC Mach field inside an ideal bell (`plot_moc_field`).



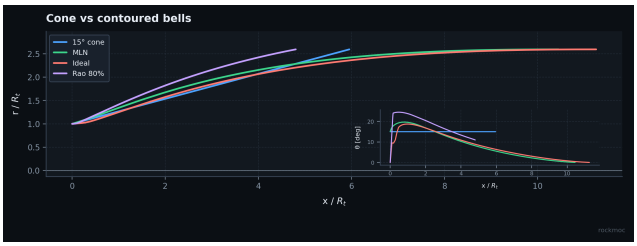
(a) MLN $C^{\pm}$ net and Mach field.



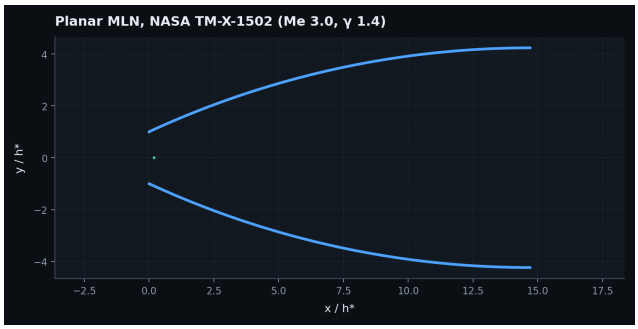
(b) Rao thrust-optimised bell field.



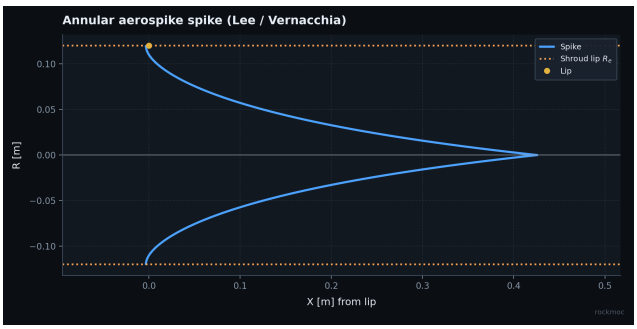
(a) Contour families at equal M_e, R_t: Rao < TIC < MLN < ideal.



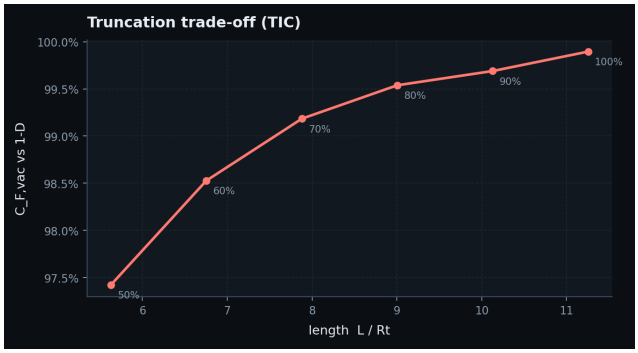
(b) 15° cone vs contoured bells (C_F/1-D cone 0.969).



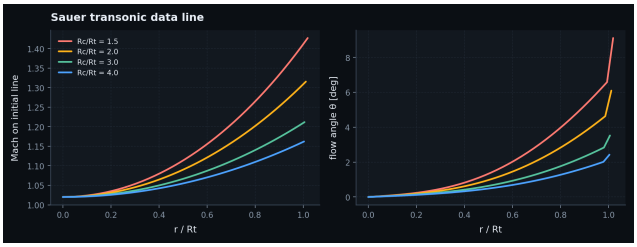
(a) Planar TM-X-1502 MLN; $\theta_{max} = \nu_e/2$ exactly.



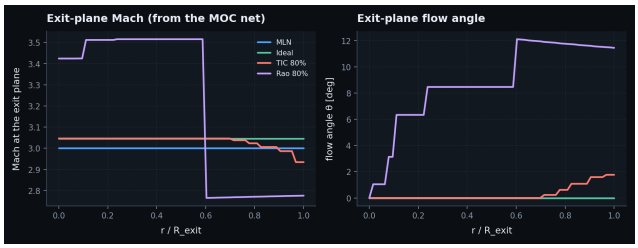
(b) Lee/Vernacchia annular aerospike spike.



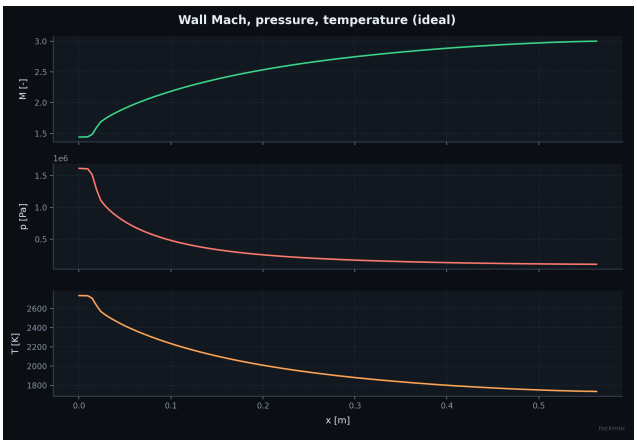
(a) TIC length vs C_F trade.



(b) Transonic initial-value line near the throat.

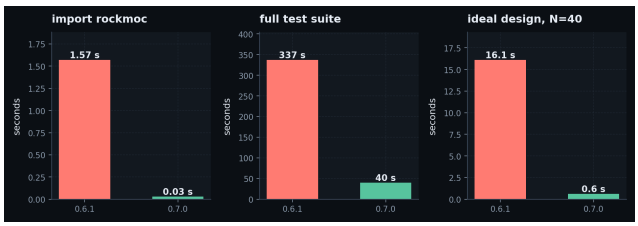


(a) Exit-plane M, θ profiles used by field C_F.

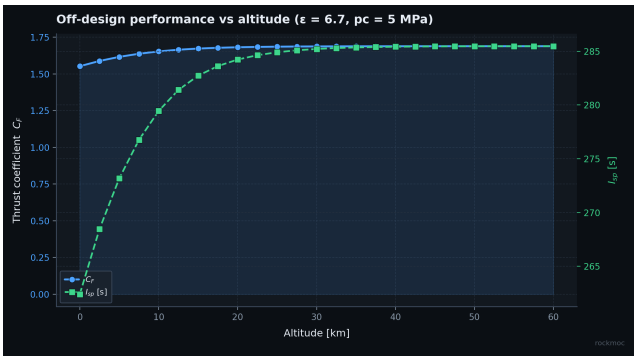


(b) Wall M, p, T along ideal contour.

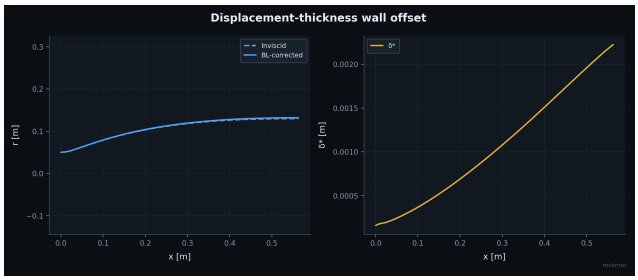
5.2 Performance, BL, analysis



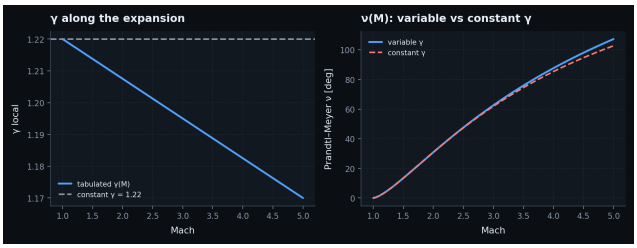
(a) C_F/Isp breakdown.



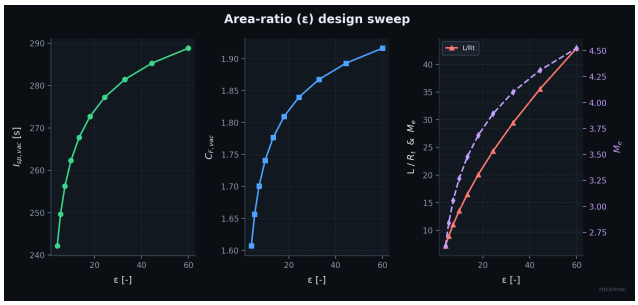
(b) C_F, Isp vs altitude.



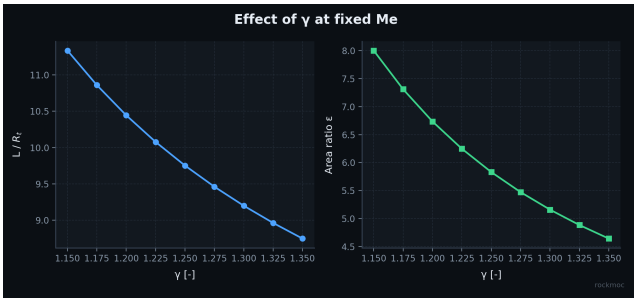
(a) Momentum-integral BL: inviscid vs corrected wall and $\delta^*(x)$.



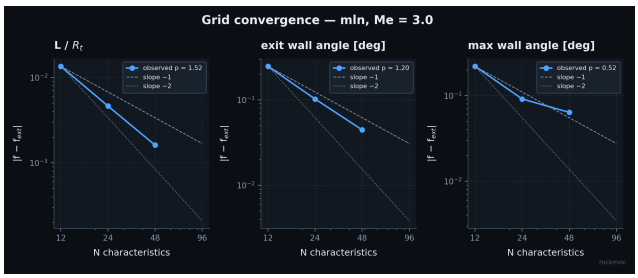
(b) Multi-γ thermally perfect isentrope vs constant γ.



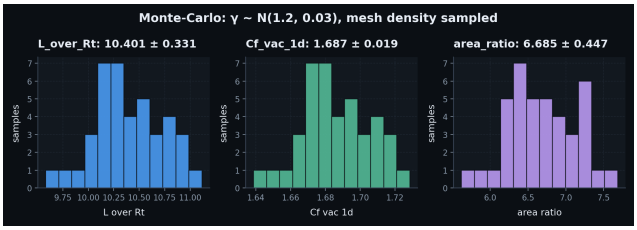
(a) Isp, C_F, L/R_t vs design ε.



(b) Effect of γ at fixed M_e.



(a) Grid convergence (observed orders: L/R_t 1.5, θ_e 1.2).



(b) Monte-Carlo UQ of MLN geometry.

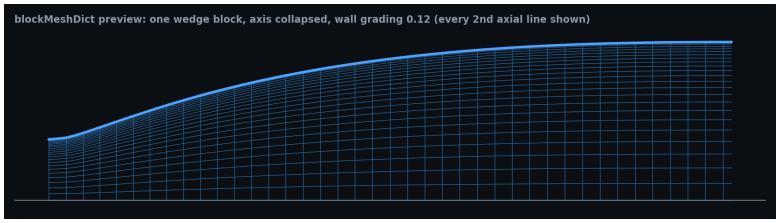
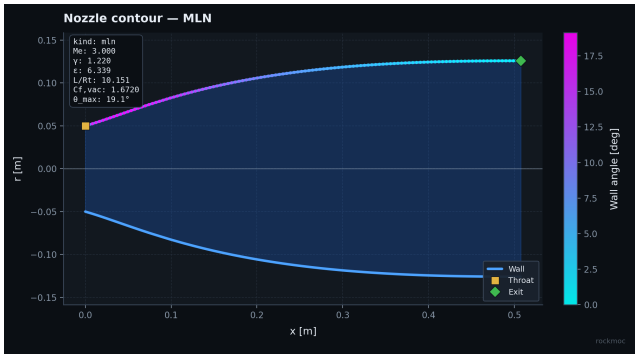


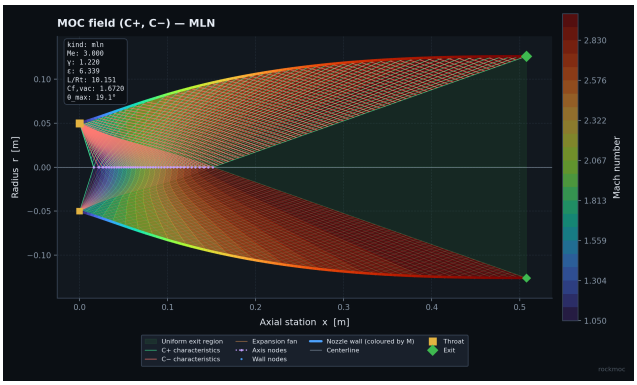
Figure 11: OpenFOAM mesh from `export_blockmesh` (blockMesh, checkMesh OK).

5.3 Per-design dashboards (`DesignResult.plot`)

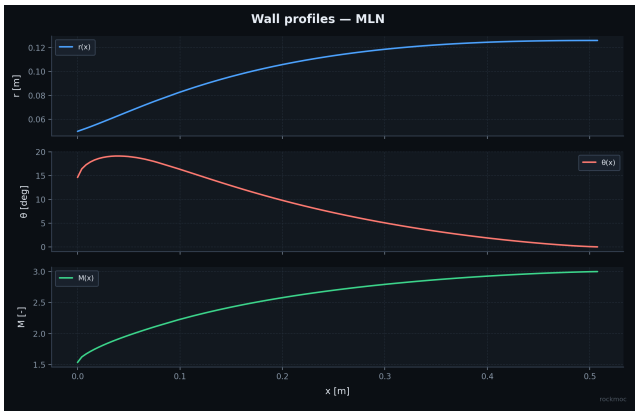
Each design writes `<kind>_contour`, `_moc_field`, `_wall_profiles`, `_mach_pressure`; $\gamma = 1.22$, $M_e = 3$, $n = 40$.



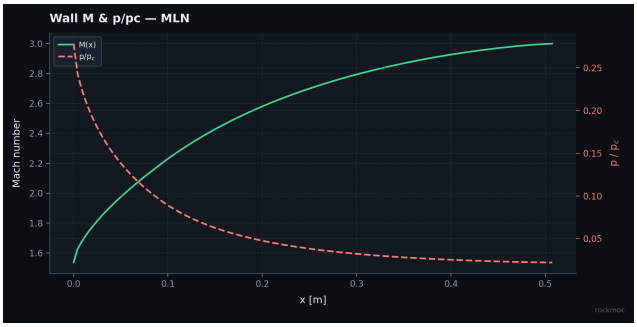
(a) MLN contour + metrics.



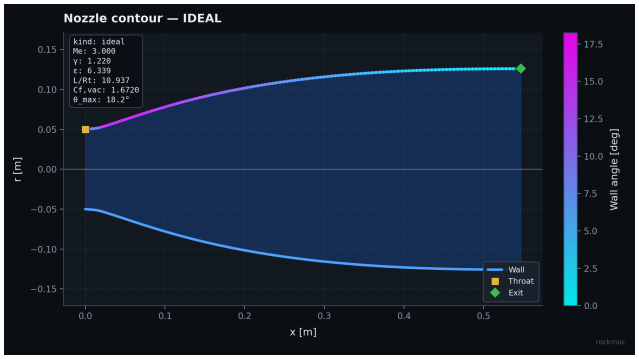
(b) MLN MOC field.



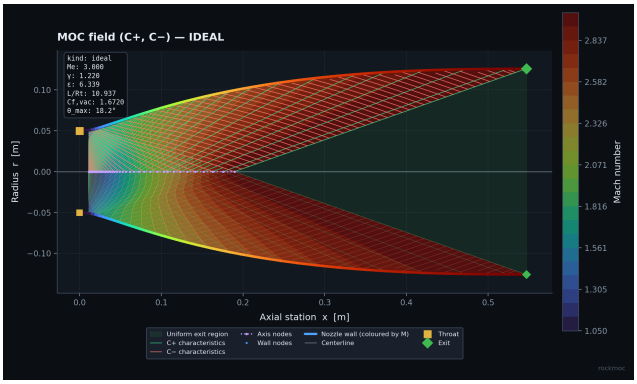
(a) MLN wall r, θ.



(b) MLN wall M, p/p_c.



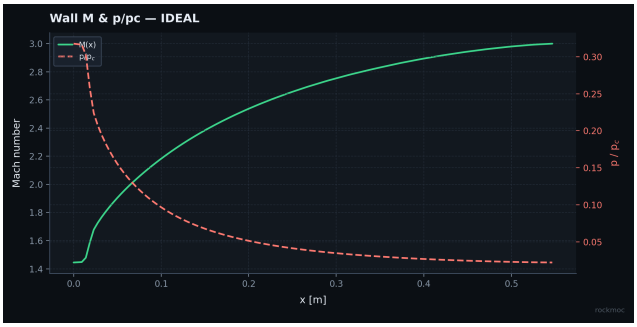
(a) Ideal contour.



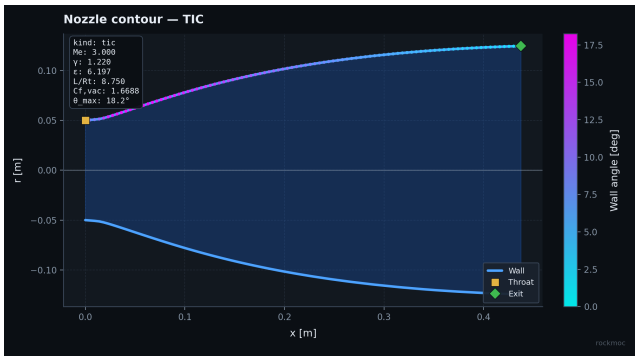
(b) Ideal MOC field.



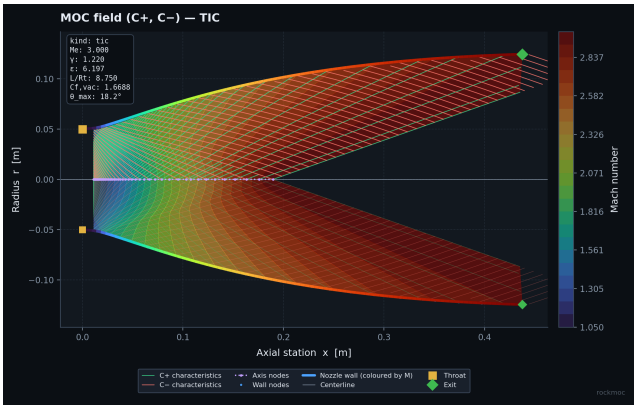
(a) Ideal wall profiles.



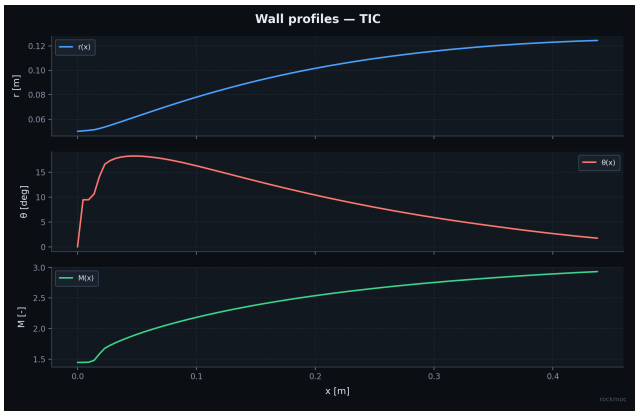
(b) Ideal wall M, p/p_c.



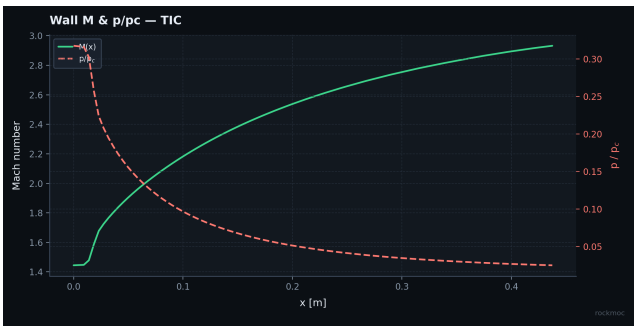
(a) TIC (80%) contour.



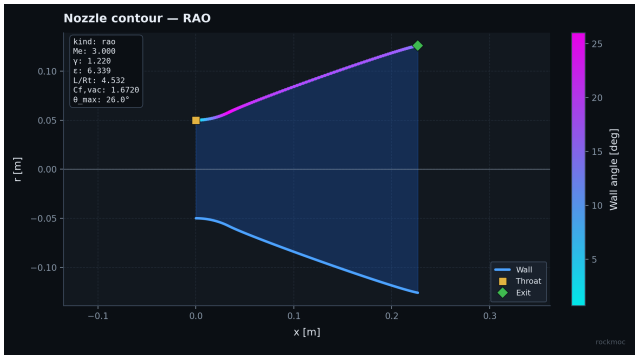
(b) TIC MOC field.



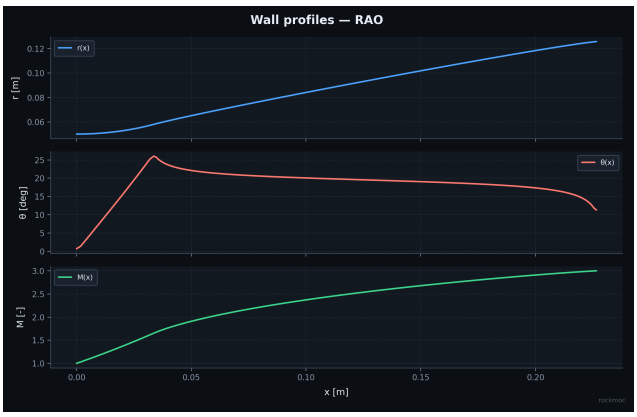
(a) TIC wall profiles.



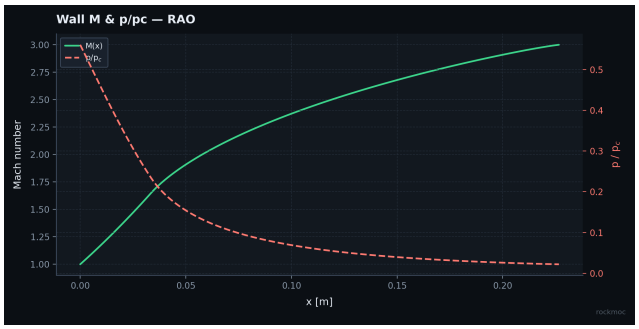
(b) TIC wall M, p/p_c.



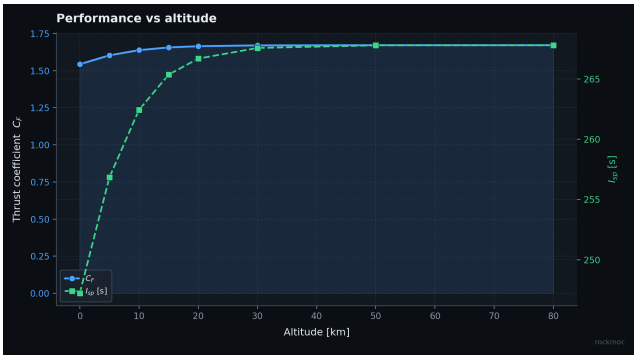
(a) Rao contour.



(b) Rao wall profiles.

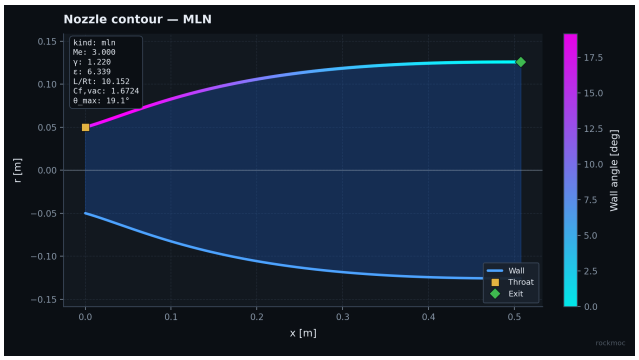


(a) Rao wall M, p/p_c.

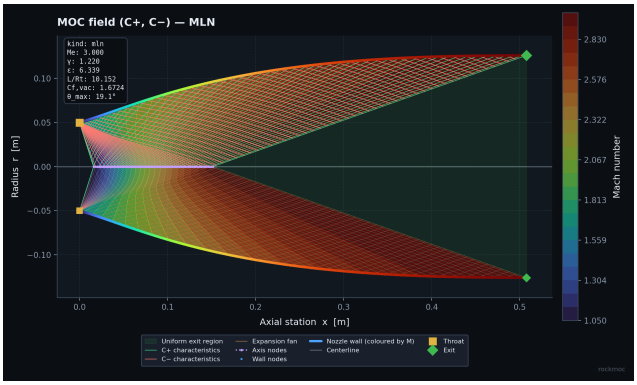


(b) MLN C_F/Isp vs altitude.

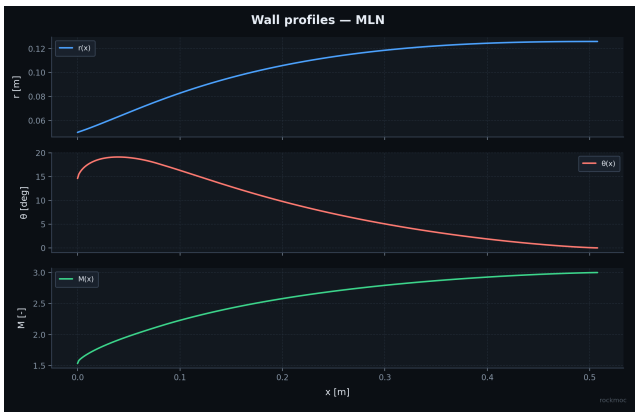
5.4 Production pipeline (`ProductionResult.plot`)



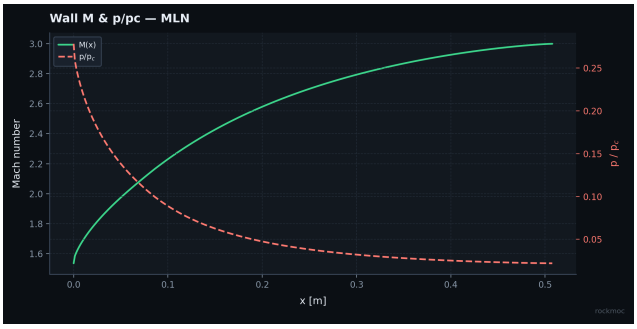
(a) Production contour ($n \geq 120$).



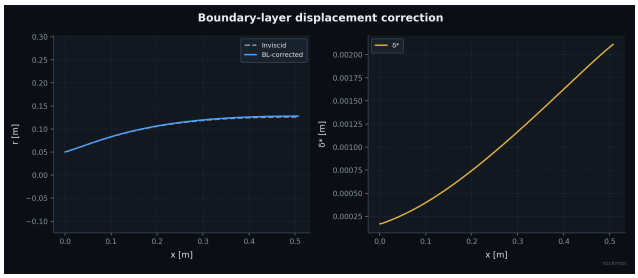
(b) Production MOC field.



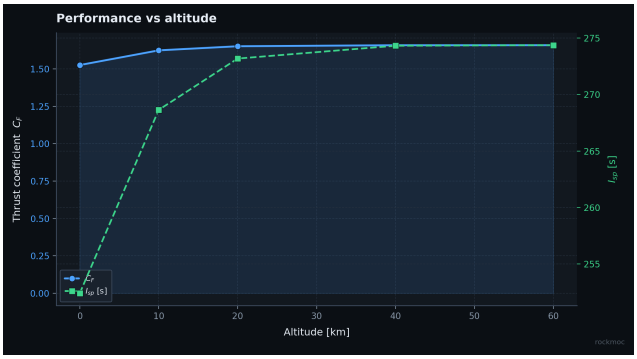
(a) Production wall profiles.



(b) Production wall M, p/p_c.

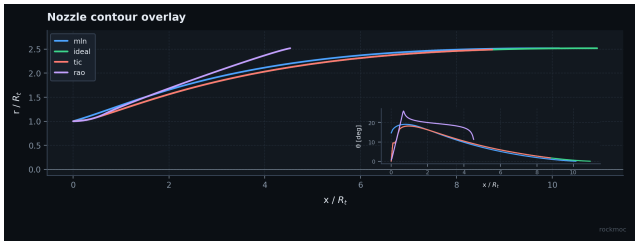


(a) BL-corrected wall and δ^*.

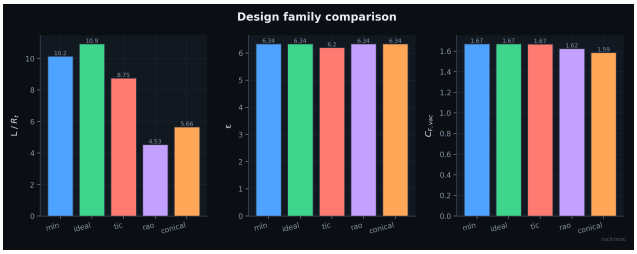


(b) Altitude performance with friction loss.

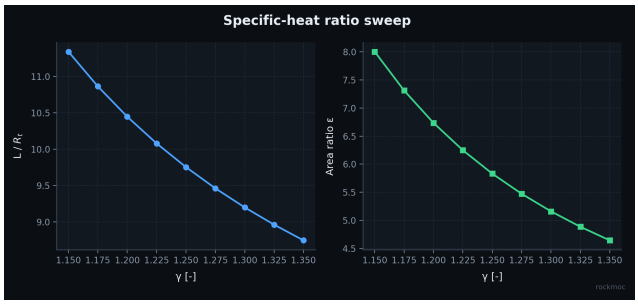
5.5 Sweeps and comparisons



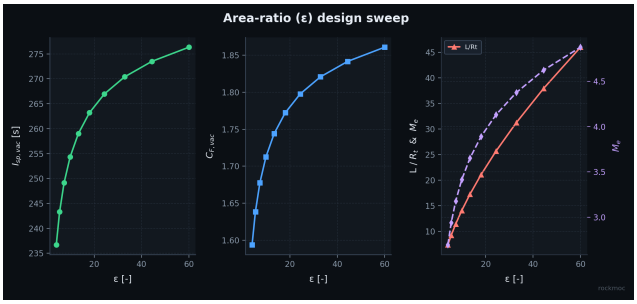
(a) Family overlay with wall-angle inset.



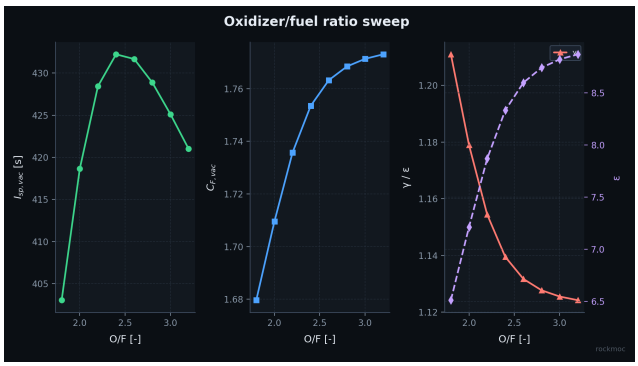
(b) Family bar chart: L/R_t, ε, C_F, η_n.



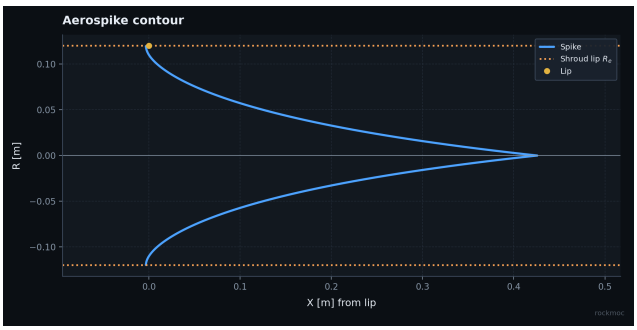
(a) γ sweep.



(b) ε sweep.

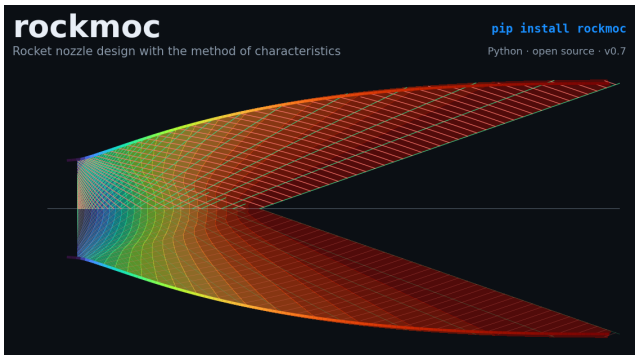


(a) LOX/RP-1 O/F sweep (CEA).



(b) Aerospike contour (dashboard).

5.6 Project cards



(a) Cover card.



(b) Families card.



(a) v0.7 physics-fix card.



(b) What's-new card.

6 Validation

- **Tests**: 113 pass, 1 skipped (`pytest`), incl. hypothesis property tests and `tests/test_accuracy_v08.py` (15): ZPG integral = correlation (10^{-4}), fallback path, cold-wall δ^*, ΔC_F 0.2–3%, Van Driest II (±3%, $M = 2, 3, 5$), Hall order > 3, multi-γ limit, coarse-lattice Rao.
- **Checks**: `run_all_validation()` 50/50.
- **Cited campaign**: `run_deep_validation()` 43/43 cases, 170 metrics, 12 sources [1–10, 12, 16]; max error 0.014%.
- **Published experiments and reference designs** (`rockmoc.validate.experimental`, `docs/validation_experimental`: 74/74 gated checks pass (v0.9 adds Zebbiche MLN, KL/Euler throat, chamber-start BL, separation, shock, kinetics and two-phase rows); 8 informational rows document known limits.

Nozzle	Reference and result
Conical BL	Boldman et al., 30°–15° cone, air, M 2. +24, +2% (Bartz −16%, Sasman–Cres −43% (inlet thermal history not model
Throat	$R_c/R_t = 2.01$ wall p/p_0: Hall −0.3% axis/wall M 0.56/1.56 vs 0.8/1.4 (info)
Conical	wall p/p_0 vs 1-D at ε 8.4/15.1/24.1: +
Rao TOC	SP-8120 80% chart ε=10–70: $\lvert\Delta\theta_N\rvert \leq 1$ whole wall within 0.37% of r_e, θ_e −0.26
Aerospike	Lee/Vernacchia vs first-principles centre exact.
MLN	exit area = isentropic A/A^* (mass) at 3 MLN table was reachable.

- **CFD hand-off**: `blockMeshDict` meshes cleanly (blockMesh/checkMesh).
- CI: Windows/Linux/macOS, Python 3.10–3.13, nightly, Numba variant.

7 Limits addressed in v0.9

Each former limit now has a model and a validation gate (`rockmoc.validate.experimental`, 74/74):

Tight throats Kliegel–Levine series $u = 1+\varepsilon u_1+\varepsilon^2(u_1+u_2)+\varepsilon^3(u_1+2u_2+u_3)$, $\varepsilon = 1/(1+R)$, for $0.4 \le R < 1.5$ (Hall's $z = x\sqrt{2R/(\gamma+1)}$ is the only scaling that keeps $\partial u/\partial r = \partial v/\partial x$ order by order); closed-form C_d. Below 0.4 a 2-D Euler solver (`moc.transonic_euler`, first-order Rusanov FV, Richardson-extrapolated). Cuffel $R = 0.625$: KL axis M 0.79, C_d 0.982; Euler 0.82, 0.982; measured 0.8, 0.985.
Boundary layer Bartz energy integral (interaction exponent 0) $d\phi/ds = St\,(T_{aw}-T_w)/(T_0-T_w)-\phi[d\ln\rho U/ds+ d\ln r/ds + d\ln(T_0 - T_w)/ds]$, two-layer 1/7 profiles over δ and Δ giving $H(M, T_w/T_0, \Delta/\delta)$ (can be negative), chamber-inlet start, and the relaminarisation switch $K_{ax} > 2.88 \times 10^{-6}$ [17].
Separation criteria from Summerfield to Stark, truncated-nozzle or Kalt–Badal recovery C_F (identical to RocketCEA), NASA TP-1207 guards.
Internal shocks direct MOC of a given wall with a cell-Jacobian crossing test: TOP parabolas and arc-throat cones flagged, ideal contours clean.
Kinetics quasi-1-D finite-rate Cantera nozzle; frozen and equilibrium limits reproduced exactly; η_{kin} 0.980/0.991/0.997 for 5 mm/5 cm/50 cm H_2/O_2 throats.
Two-phase gas–particle velocity/thermal lag; no-lag limit equals the mixture isentrope.
MLN data Zebbiche's perfect-gas axisymmetric MLN table: L/r^* within 0.03%, angles within 0.02°.

8 Limits and recommended use

- Inviscid core outside the throat; separation and internal shocks are detected/estimated, not resolved (no shock fitting, no restricted-shock-separation pattern).
- BL: integral method; cooled-throat $\delta^* \approx 0$ is sign-sensitive; no film cooling.
- KL start line is vertical; at $R < 0.6$ its wall Mach rises steeply downstream, use the Euler solver there.
- Kinetics: reaction mechanism supplied by the user (H_2/O_2 default); equilibrium start at $\varepsilon = 1.05$. Two-phase: particle size is an input; starts in equilibrium at $\varepsilon = 1.05$ (slightly optimistic).
- Separation criteria scatter by ∼20%; report Schmucker and Stark as a band.

Workflow: choose family → `design_*` at $n \approx 40$ → check convergence → `design_production` with CEA γ, wall T, η_{chem} → export DXF/OpenFOAM → CFD verification.

9 Changelog

v0.9 (unreleased): Kliegel–Levine throat and C_d, 2-D Euler throat solver, Bartz two-thickness BL with chamber start and relaminarisation, separation criteria, internal-shock detector, Cantera finite-rate kinetics, two-phase lag, Zebbiche MLN validation. **v0.8**: integral BL + fallback, Bartz transport, wall temperature, friction ΔC_F and loss stack, kinetic bounds, Hall throat, thermally perfect multi-γ, Rao TOC singularity fix, `website/make_assets.py`. **v0.7.0**: C^- compatibility sign fix, rebuilt axisymmetric MLN and ideal lattice, Rao TOC by MOC, field C_F, Numba kernels, exports (DXF, OpenFOAM), TOML + argparse CLI, CI.

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